

MLOps Assignment: DistilBERT Text Classification Pipeline

Overview

This project demonstrates a complete MLOps workflow using Hugging Face Transformers, Kaggle GPU training, Weights & Biases (W&B) experiment tracking, and Hugging Face Hub deployment. The model used in this assignment is `distilbert-base-cased`, which is a lightweight and faster version of BERT suitable for text classification tasks.

The workflow includes:

- Loading a pre-trained DistilBERT model from Hugging Face
 - Fine-tuning the model on a classification dataset using Kaggle GPU
 - Tracking experiments and metrics with W&B
 - Evaluating model performance using Accuracy and F1 Score
 - Saving evaluation reports as W&B artifacts
 - Uploading the trained model and tokenizer to Hugging Face Hub
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Task 2: Load a Pre-trained Model from Hugging Face

Description

This script loads the `distilbert-base-cased` tokenizer and model from Hugging Face Transformers. The model is configured for binary classification with 2 output labels.

Key Features

- Uses `DistilBertTokenizerFast`
- Loads `DistilBertForSequenceClassification`
- Runs on GPU using CUDA
- Supports maximum sequence length of 512

Model Details

- Model Name: `distilbert-base-cased`
- Output Labels: `2`
- Device: `CUDA GPU`

Output

The script prints:

- Tokenizer loading confirmation
 - Model loading confirmation
 - Active device information
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Task 2: Train the Model on Kaggle & Track with W&B

Description

This script fine-tunes the DistilBERT model using Hugging Face Trainer API inside a Kaggle Notebook with GPU acceleration. Weights & Biases (W&B) is integrated for experiment tracking.

Features

- W&B experiment tracking
- Accuracy and weighted F1-score computation
- GPU training on Kaggle
- Automatic evaluation after training
- Hyperparameter logging

Training Configuration

Parameter	Value
Epochs	3
Train Batch Size	16
Eval Batch Size	32
Learning Rate	3e-5
Weight Decay	0.01
Logging Steps	50

W&B Tracking

The following metrics are logged:

- Training Loss
 - Validation Loss
 - Accuracy
 - F1 Score
 - Hyperparameters
 - Hugging Face Model URL
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Task 3: Evaluate & Save Results

Description

After training, the model is evaluated on the test dataset.

Evaluation Metrics

- Accuracy
- F1 Score
- Evaluation Loss

Additional Features

- Generates a detailed classification report
- Saves report as `eval_report.json`
- Uploads evaluation report to W&B as an artifact

Artifact Logging

The evaluation report is versioned and stored in W&B for reproducibility and experiment tracking.

Task 4: Save Model to Hugging Face Hub

Description

The trained DistilBERT model and tokenizer are uploaded to Hugging Face Hub for public access.

Hugging Face Repository

Model Repository:

<https://huggingface.co/Vishnu1010/distilbert-imdb-classifier>

Uploaded Components

- Fine-tuned DistilBERT model
 - Tokenizer files
 - Configuration files
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GitHub Repository

GitHub Link:

<https://github.com/justCodeItVishnu/mlOps>

W&B Dashboard

Weights & Biases Dashboard:

<https://wandb.ai/g25ait2128-prom-iit-rajasthan/mlops-assignment2?nw=nwuserg25ait2128>

Requirements

Install the required libraries using:

```
pip install transformers datasets torch wandb scikit-learn huggingface_hub
```

Running the Project

Step 1: Load Model

Run the model loading script.

Step 2: Train the Model

Run the Kaggle training script with GPU enabled.

Step 3: Evaluate

The script automatically evaluates the model after training.

Step 4: Upload Model

The trained model and tokenizer are pushed to Hugging Face Hub.

Results

Metric	Score
Accuracy	Logged in W&B
F1 Score	Logged in W&B
Eval Loss	Logged in W&B

Conclusion

This assignment demonstrates an end-to-end MLOps pipeline using Hugging Face Transformers, Kaggle GPU infrastructure, W&B experiment tracking, and Hugging Face Hub deployment. The workflow ensures reproducibility, model monitoring, evaluation tracking, and public model sharing in a production-style setup.